ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Sequences

Read the trigger, name the data, choose the construction, write the bottom line.

SEQUENCES · CH3 · L1

ENG. ESLAM AHMED

The Map

This lesson is not a list of formulas. It is six exam moves. The question starts when you recognise the words that tell you what the working must become.

01 The rule (nth term)
TRIGGER *"nth term", "expression in terms of n", "the rule"*

02 A named term or membership test
TRIGGER *"20th term", "is 150 a term?", "which term is ...?"*

03 The sum
TRIGGER *"sum", "total", "altogether", "from the 50th to the 100th"*

04 Two clues $\to$ a and d
TRIGGER *two facts about the same sequence, for example* u_{20} *and* S_{30}

05 Letters inside first terms
TRIGGER *terms like* $8p,\ 7p-3,\ 4p+2$

06 Context and geometry
TRIGGER *savings, shelves, polygons, counterexamples*

BANK EVIDENCE 30 unique solved source questions were mined from the Sequences bank, worth 134 marks. The largest weight is arithmetic series and algebraic sequence terms.

01 The Rule — nth Term

TRIGGER *"nth term", "expression in terms of n", first few terms*

BECOMES A rule that jumps straight from the position n to the term u_n.

FIRST LINE TO WRITE

$$d = \text{second term} - \text{first term} \qquad \text{then} \qquad u_n = a + (n-1)d$$

SIMPLEST STRATEGY

1. Read the first term as a.
2. Find the common difference d.
3. Write $u_n = a + (n-1)d$.
4. Expand and tidy.
5. Check by putting $n = 1$.

WORKED MODEL

First four terms: $85,\ 79,\ 73,\ 67.$

$$d = 79 - 85 = -6$$

$$u_n = 85 + (n-1)(-6) = 85 - 6n + 6 = 91 - 6n$$

BOTTOM LINE

$\boxed{u_n = 91 - 6n}$.

Practice 01

SAME IDEA Find the nth-term rule from the first few terms.

QUESTION

> Here are the first four terms of an arithmetic sequence:
>
> $$38, \quad 31, \quad 24, \quad 17$$
>
> Find an expression, in terms of n, for the nth term of the sequence.

YOUR SOLUTION

02 A Named Term or a Membership Test

TRIGGER *"the 20th term", "which term is ...?", "is 150 a term?"*

BECOMES Either substitute a position into the rule, or solve the rule for n.

FIRST LINE TO WRITE

> named term: substitute n membership: solve $u_n =$ given value

SIMPLEST STRATEGY

1. If the question says "20th", put $n = 20$.
2. If it asks whether a value is in the sequence, set u_n equal to that value.
3. Solve for n.
4. The value is a term only if n is a positive whole number.

WORKED MODEL

> *The nth term is* $4n + 7$. *Show a counterexample to the rule* $5n + 4$.
>
> $$5(3) + 4 = 19$$
>
> But 19 also appears in the first sequence:
>
> $$4(3) + 7 = 19$$

BOTTOM LINE

A suitable counterexample is $\boxed{19}$.

Practice 02

SAME IDEA Substitute a position, or solve the rule for n.

QUESTION

> The nth term of a sequence is $5n + 17$.
>
> (a) Find the 12th term.
>
> (b) Is 152 a term in this sequence? Show your working.

YOUR SOLUTION

03 The Sum — Arithmetic Series

TRIGGER *"sum", "total", "altogether", S_n, "from term r to term s"*

BECOMES A series question. You are adding terms, not finding one term.

FIRST LINE TO WRITE

$$S_n = \frac{n}{2}(2a + (n-1)d) \qquad \text{or} \qquad S = \frac{\text{number}}{2}(\text{first} + \text{last})$$

SIMPLEST STRATEGY

1. Decide whether it is first-n or a block from one position to another.
2. For a block, find the first term of the block and the last term of the block.
3. Count the terms carefully: last position minus first position plus 1.
4. Use the first-plus-last version when both ends of the block are known.

WORKED MODEL

First terms: 8, 15, 22, 29, 36. *Find the sum from the 50th term to the 100th term inclusive.*

$$d = 7, \qquad u_n = 8 + 7(n-1) = 7n + 1$$

$$u_{50} = 351, \qquad u_{100} = 701, \qquad \text{number} = 100 - 50 + 1 = 51$$

$$S = \frac{51}{2}(351 + 701) = 26826$$

BOTTOM LINE

$\boxed{26826}$.

Practice 03

SAME IDEA Add a block of terms by finding both ends and counting carefully.

QUESTION

> An arithmetic sequence has first term 1 and common difference 4. Find the sum of all the terms from the 41st term to the 100th term inclusive.

YOUR SOLUTION

04 Two Clues $\to a$ and d

TRIGGER *two facts about the same arithmetic sequence, for example* $u_{20} = 123$ *and* $S_{30} = 2880$

BECOMES Two simultaneous equations in a and d.

FIRST LINE TO WRITE

$$u_r = a + (r-1)d \qquad \text{and} \qquad S_n = \frac{n}{2}(2a + (n-1)d)$$

SIMPLEST STRATEGY

1. Turn the term fact into equation 1.
2. Turn the sum fact into equation 2.
3. Simplify both equations.
4. Eliminate one letter.
5. Write both values before answering the final request.

WORKED MODEL

$$u_{20} = a + 19d = 123 \qquad (1)$$
$$S_{30} = \frac{30}{2}(2a + 29d) = 2880$$
$$2a + 29d = 192 \qquad (2)$$

Double (1), then subtract (2):

$$2a + 38d = 246 \qquad 9d = 54 \qquad d = 6$$

$$a + 19(6) = 123 \qquad a = 9$$

BOTTOM LINE

$\boxed{a = 9,\ d = 6}$.

Practice 04

SAME IDEA Turn two facts into two equations, then solve.

QUESTION

> An arithmetic series has first term a and common difference d. The 21st term is 109. The sum of the first 52 terms is 4381. Work out the 5th term of the series.

YOUR SOLUTION

05 Letters Inside the First Terms

TRIGGER *terms such as* $8p,\ 7p-3,\ 4p+2$

BECOMES An equal-differences question first. The real sequence appears after the letter is found.

FIRST LINE TO WRITE

$$\text{term 2} - \text{term 1} = \text{term 3} - \text{term 2}$$

SIMPLEST STRATEGY

1 Use equal differences to solve the letter.

2 Replace the letter in the original terms.

3 Now identify a and d.

4 Continue with u_n or S_n, depending on what the question asks.

WORKED MODEL

The first three terms are $8p,\ 7p-3,\ 4p+2.$

$$(7p-3) - 8p = (4p+2) - (7p-3)$$

$$-p-3 = -3p+5, \qquad 2p = 8, \qquad p = 4$$

So the sequence is:

$$32, \quad 25, \quad 18, \ldots \qquad a = 32, \quad d = -7$$

If $S_n = -1914$,

$$\frac{n}{2}(2(32) + (n-1)(-7)) = -1914$$

$$\frac{n}{2}(71 - 7n) = -1914 \qquad \Rightarrow \qquad 7n^2 - 71n - 3828 = 0$$

$$n = \frac{71 + 335}{14} = 29$$

BOTTOM LINE

$\boxed{n = 29}$.

Practice 05

SAME IDEA Use equal differences first, then continue with the real sequence.

QUESTION

> The first three terms of an arithmetic sequence are
>
> $$5q, \quad 4q+3, \quad q+12.$$
>
> Find q. Then find the sum of the first 20 terms of the sequence.

YOUR SOLUTION

06 Context and Geometry

TRIGGER *savings by year, shelves, polygon angles, or one counterexample*

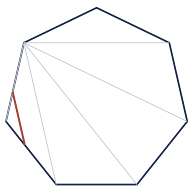
BECOMES A hidden sequence. Translate the story first, then use the same sequence rules.

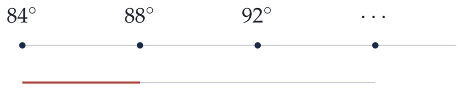
FIRST LINE TO WRITE

> name a, d, n then write the outside rule beside the sequence rule

SIMPLEST STRATEGY

1. Savings: count years first, especially when the years are inclusive.
2. Shelves or patterns: count each object type in terms of n.
3. Polygon angles: sequence sum equals $180(n-2)$.
4. Counterexample: one valid value that breaks the claim is enough.





$$S_n = \frac{n}{2}(2a + (n-1)d)$$
$$S_n = 180(n-2)$$

sequence of interior angles

FIGURE Match the angle-series sum with the polygon interior-angle sum.

WORKED MODEL

The interior angles of a polygon form an arithmetic sequence. The smallest angle is 84°, and the common difference is 4°.

$$S_n = \frac{n}{2}(2(84) + (n-1)4) = 2n^2 + 82n$$

The polygon interior sum is:

$$180(n-2)$$

Set equal:

$$2n^2 + 82n = 180(n-2)$$

$$n^2 - 49n + 180 = 0 \qquad (n-45)(n-4) = 0$$

Since $n > 5$, $n = 45$, so

$$\text{sum} = 180(45-2) = 7740$$

BOTTOM LINE

$\boxed{7740°}$.

Practice 06

SAME IDEA Translate the story into a sequence rule and an outside rule.

QUESTION

> The interior angles of a polygon form an arithmetic sequence. The smallest angle is $80°$, and the common difference is $5°$. The polygon has more than 5 sides. Work out the sum of the interior angles of the polygon.

YOUR SOLUTION

Quick Reference

TRIGGER → ACTION

IF YOU SEE	DO THIS FIRST
"nth term", "expression in terms of n"	$u_n = a + (n-1)d$.
"the 20th term"	Substitute $n = 20$.
"is this value a term?"	Solve $u_n =$ value and check n is a positive integer.
"sum", "total", "altogether"	$S_n = \frac{n}{2}(2a + (n-1)d)$.
"from term r to term s"	number of terms $= s - r + 1$.
two facts about the sequence	write two equations, then solve for a, d.
letters in first terms	equal differences first.
polygon angles	sequence sum $= 180(n-2)$.

THE BOTTOM LINES

1. **Rule:** $u_n = 3n - 2$, or another clean expression in n.
2. **Term:** one value, such as $u_{20} = 77$.
3. **Membership:** the position n, or not a term.
4. **Sum:** one total, with units if the problem has units.
5. **Two clues:** a and d, then the exact thing asked for.
6. **Context:** the story answer, not only the algebra.

FORMULA BANK

$u_n = a + (n-1)d$ $\quad d = u_2 - u_1$ $\quad S_n = \frac{n}{2}(2a + (n-1)d)$ $\quad S = \frac{\text{number}}{2}(\text{first} + \text{last})$ $\quad 180(n-2)$

PRACTICE ORDER

Linear nth term → block sums → two-equation questions → letters/context/geometry.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Coordinate Geometry

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

COORDINATE GEOMETRY · CH3 · L2

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The Map

This lesson collapses Coordinate Geometry into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use midpoint

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use gradient

TRIGGER *"gradient", "tangent", "rate of change"*

03 Find equation of a line

TRIGGER *"work out", "find", a missing value*

04 Check if a point lies on a line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 60 website questions, 60 checked solutions, 3 local PDFs read.

01 Use midpoint

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{5+13}{2}, \frac{-4+1}{2}\right)$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q1

Find the midpoint

$$\left(\frac{5+13}{2}, \frac{-4+1}{2}\right)$$

(b)

$$y = 2 - 3x$$

Finish: $(9, -1.5)$, gradient -3, and no.

FORMULA BANK

Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$. Gradient $= \frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 1 The point A has coordinates (5, -4) The point B has coordinates (13, 1) (a) Work out the coordinates of the midpoint of AB. (,) (2) Line L has equation $y = 2 - 3x$ (b) Write down the gradient of line L. (1) Line L has equation $y = 2 - 3x$ (c) Does the point with coordinates (100, -302) lie on line L? You must give a reason for your answer. (1)

YOUR SOLUTION

02 Use gradient

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{5+13}{2}, \frac{-4+1}{2}\right)$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q1

Find the midpoint

$$\left(\frac{5+13}{2}, \frac{-4+1}{2}\right)$$

(b)

$$y = 2 - 3x$$

Finish: $(9, -1.5)$, gradient -3, and no.

FORMULA BANK

Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$. Gradient $= \frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 1 The point A has coordinates (5, -4) The point B has coordinates (13, 1) (a) Work out the coordinates of the midpoint of AB. (,) (2) Line L has equation $y = 2 - 3x$ (b) Write down the gradient of line L. (1) Line L has equation $y = 2 - 3x$ (c) Does the point with coordinates (100, -302) lie on line L? You must give a reason for your answer. (1)

YOUR SOLUTION

03 Find equation of a line

TRIGGER *"work out", "find", a missing value*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{5+13}{2}, \frac{-4+1}{2}\right)$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q1

Find the midpoint

$$\left(\frac{5+13}{2}, \frac{-4+1}{2}\right)$$

(b)

$$y = 2 - 3x$$

Finish: $(9, -1.5)$, gradient -3, and no.

FORMULA BANK

Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$. Gradient $= \frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 1 The point A has coordinates (5, -4) The point B has coordinates (13, 1) (a) Work out the coordinates of the midpoint of AB. (,) (2) Line L has equation $y = 2 - 3x$ (b) Write down the gradient of line L. (1) Line L has equation $y = 2 - 3x$ (c) Does the point with coordinates (100, -302) lie on line L? You must give a reason for your answer. (1)

YOUR SOLUTION

04 Check if a point lies on a line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\left(\frac{5+13}{2}, \frac{-4+1}{2}\right)$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q1

Find the midpoint

$$\left(\frac{5+13}{2}, \frac{-4+1}{2}\right)$$

(b)

$$y = 2 - 3x$$

Finish: $(9, -1.5)$, gradient -3, and no.

FORMULA BANK

Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$. Gradient $= \frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 1 The point A has coordinates (5, -4) The point B has coordinates (13, 1) (a) Work out the coordinates of the midpoint of AB. (,) (2) Line L has equation y = 2 - 3x (b) Write down the gradient of line L. (1) Line L has equation y = 2 - 3x (c) Does the point with coordinates (100, -302) lie on line L? You must give a reason for your answer. (1)

YOUR SOLUTION

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Coordinate Geometry question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$M = \left(\frac{j+6}{2}, \frac{15+k}{2}\right)$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q22

Find the midpoint

$$M = \left(\frac{j+6}{2}, \frac{15+k}{2}\right)$$

Find the gradient

$$\frac{\frac{15+k}{2} - 1}{\frac{j+6}{2} + 4} = 2$$

Finish: $j = -2,\ k = 11$

FORMULA BANK

Midpoint $= \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$. Gradient $= \frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 22 Triangle HJK is isosceles with HJ = HK and JK = 80 H is the point with coordinates (-4, 1) J is the point with coordinates (j, 15) where $j < 0$ K is the point with coordinates (6, k) M is the midpoint of JK. The gradient of HM is 2 Find the value of j and the value of k. j = k =

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use midpoint
"gradient", "tangent", "rate of change"	Use gradient
"work out", "find", a missing value	Find equation of a line
"work out", "show", "hence", a familiar exam phrase	Check if a point lies on a line
"work out", "find", a missing value	Find the gradient

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Functions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FUNCTIONS · CH3 · L3

ENG. ESLAM AHMED

The Map

This lesson collapses Functions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Substitute into a function

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find an inverse

TRIGGER *"work out", "find", a missing value*

03 Use composite functions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Solve from function notation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Evaluate fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 31 website questions, 31 checked solutions, 3 local PDFs read.

01 Substitute into a function

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = x^2 - 2x$$

SIMPLEST STRATEGY

1 Underline the target: what is the question asking you to find?

2 Write the rule, theorem, or equation before any calculator work.

3 Substitute one value at a time and keep units/unknowns visible.

4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q21

Calculate value

$$f(x) = x^2 - 2x$$

Let

$$y = (x+2)^2 - 1$$

Finish: $h^{-1}(x) = \sqrt{x+1} - 2$

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y, composite means work inside first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 21 The functions f and g are such that f(x) = x2 - 2x g(x) = x + 3 The function h is such that h(x) = fg(x) for x -2 Express the inverse function h-1(x) in the form h-1(x) = ... h-1(x) =

YOUR SOLUTION

02 Find an inverse

TRIGGER *"work out", "find", a missing value*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = x^2 - 2x$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q21

Calculate value

$$f(x) = x^2 - 2x$$

Let

$$y = (x+2)^2 - 1$$

Finish: $h^{-1}(x) = \sqrt{x+1} - 2$

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y, composite means work inside first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 21 The functions f and g are such that f(x) = x2 - 2x g(x) = x + 3 The function h is such that h(x) = fg(x) for x -2 Express the inverse function h-1(x) in the form h-1(x) = ... h-1(x) =

YOUR SOLUTION

03 Use composite functions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = x^2 - 2x$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q21

Calculate value

$$f(x) = x^2 - 2x$$

Let

$$y = (x+2)^2 - 1$$

Finish: $h^{-1}(x) = \sqrt{x+1} - 2$

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y, composite means work inside first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 21 The functions f and g are such that f(x) = x2 - 2x g(x) = x + 3 The function h is such that h(x) = fg(x) for x -2 Express the inverse function h-1(x) in the form h-1(x) = ... h-1(x) =

YOUR SOLUTION

04 Solve from function notation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = x^2 + 6, \qquad g(x) = x - 10$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q17

Solve quadratic equation

$$f(x) = x^2 + 6, \qquad g(x) = x - 10$$

(a)

$$g(3) = 3 - 10 = -7$$

Finish: $fg(3) = 55$, $x = 5$, excluded value 0, and $h^{-1}(x) = \dfrac{4}{2-x}$.

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y, composite means work inside first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 17 The functions f and g are defined as f(x) = x 2 + 6 g(x) = x - 10 (a) Find fg(3) (2) (b) Solve the equation fg(x) = f(x) Show clear algebraic working. (3) The function h is defined as h(x) = 2 4 x x - (c) State the value of x that cannot be included in the domain of h (1) (d) Express the inverse function h -1 in the form h -1(x) = ... h -1(x) = (3)

YOUR SOLUTION

05 Evaluate fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$f(x) = \frac{2x}{x-6}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q14

Evaluate fraction

$$f(x) = \frac{2x}{x-6}$$

(a)

$$f(10) = \frac{2(10)}{10-6} = \frac{20}{4} = 5$$

Finish: $f(10) = 5$, and $f^{-1}(x) = \dfrac{6x}{x-2}$.

FORMULA BANK

Substitution for $f(x)$, inverse by swapping x, y, composite means work inside first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 14 The function f is defined as f : x 2 6 x x - x != 6 (a) Find f(10) (1) (b) Express the inverse function f -1 in the form f -1 : x ... f -1 : x (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Substitute into a function
"work out", "find", a missing value	Find an inverse
"work out", "show", "hence", a familiar exam phrase	Use composite functions
"work out", "show", "hence", a familiar exam phrase	Solve from function notation
"work out", "show", "hence", a familiar exam phrase	Evaluate fraction

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Linear Graphs (y = mx + c)

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

LINEAR GRAPHS (Y = MX + C) · CH3 · L4

ENG. ESLAM AHMED

The Map

This lesson collapses Linear Graphs (y = mx + c) into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Read gradient and intercept

TRIGGER *"gradient", "tangent", "rate of change"*

02 Find equation from two points

TRIGGER *"work out", "find", a missing value*

03 Use parallel/perpendicular gradients

TRIGGER *"gradient", "tangent", "rate of change"*

04 Sketch from equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 8 website questions, 8 checked solutions, 3 local PDFs read.

01 Read gradient and intercept

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Linear Graphs (y = mx + c) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5y + 2x = 7$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

> **Source pattern:** Jun 2024 P2H Q18
>
> **The line is**
>
> $$5y + 2x = 7$$
>
> **Find the gradient**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **Finish:** $\frac{5}{2}$.

FORMULA BANK

$y = mx + c$. Gradient is m, intercept is c. Parallel lines have equal gradients.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> A line has gradient 3 and passes through $(2, 7)$. Find its equation.

YOUR SOLUTION

02 Find equation from two points

TRIGGER *"work out", "find", a missing value*

BECOMES A Linear Graphs (y = mx + c) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{0-(-3)}{2-0}=\frac{3}{2}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2023 P2H Q9

Solve equation
This is a reasoning step: write one clear sentence, then continue the calculation.

The gradient is

$$\frac{0-(-3)}{2-0}=\frac{3}{2}$$

Finish: $y=\frac{3}{2}x-3$.

FORMULA BANK

$y=mx+c$. Gradient is m, intercept is c. Parallel lines have equal gradients.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

A line has gradient 3 and passes through $(2, 7)$. Find its equation.

YOUR SOLUTION

03 Use parallel/perpendicular gradients

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Linear Graphs (y = mx + c) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5y + 2x = 7$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2024 P2H Q18

The line is

$$5y + 2x = 7$$

Find the gradient
This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $\frac{5}{2}$.

FORMULA BANK

$y = mx + c$. Gradient is m, intercept is c. Parallel lines have equal gradients.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> A line has gradient 3 and passes through $(2, 7)$. Find its equation.

YOUR SOLUTION

04 Sketch from equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Linear Graphs (y = mx + c) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$c = -1$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q10

Solve equation

$$c = -1$$

Find the gradient

$$\frac{2-(-1)}{2-0} = \frac{3}{2}$$

Finish: $y = \frac{3}{2}x - 1$.

FORMULA BANK

$y = mx + c$. Gradient is m, intercept is c. Parallel lines have equal gradients.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> A line has gradient 3 and passes through $(2, 7)$. Find its equation.

YOUR SOLUTION

05 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Linear Graphs (y = mx + c) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$c = -1$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q10

Solve equation

$$c = -1$$

Find the gradient

$$\frac{2-(-1)}{2-0} = \frac{3}{2}$$

Finish: $y = \frac{3}{2}x - 1$.

FORMULA BANK

$y = mx + c$. Gradient is m, intercept is c. Parallel lines have equal gradients.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> A line has gradient 3 and passes through $(2, 7)$. Find its equation.

YOUR SOLUTION

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Linear Graphs (y = mx + c) question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$c = -1$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q10

Solve equation

$$c = -1$$

Find the gradient

$$\frac{2-(-1)}{2-0} = \frac{3}{2}$$

Finish: $y = \frac{3}{2}x - 1$.

FORMULA BANK

$y = mx + c$. Gradient is m, intercept is c. Parallel lines have equal gradients.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

A line has gradient 3 and passes through $(2, 7)$. Find its equation.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"gradient", "tangent", "rate of change"	Read gradient and intercept
"work out", "find", a missing value	Find equation from two points
"gradient", "tangent", "rate of change"	Use parallel/perpendicular gradients
"work out", "show", "hence", a familiar exam phrase	Sketch from equation
"work out", "show", "hence", a familiar exam phrase	Solve equation
"work out", "find", a missing value	Find the gradient

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Graphs of Functions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

GRAPHS OF FUNCTIONS · CH3 · L5

ENG. ESLAM AHMED

The Map

This lesson collapses Graphs of Functions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Complete a table of values

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Plot the curve

TRIGGER *"draw", "plot", "construct", a diagram/grid*

03 Read roots and intersections

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Transform the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 34 website questions, 34 checked solutions, 3 local PDFs read.

01 Complete a table of values

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = x^2 - \frac{x}{2} - 3$$

SIMPLEST STRATEGY

1 Underline the target: what is the question asking you to find?
2 Write the rule, theorem, or equation before any calculator work.
3 Substitute one value at a time and keep units/unknowns visible.
4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q15

Solve quadratic equation

$$y = x^2 - \frac{x}{2} - 3$$

The completed table is

x	-3	-2	-1	0	1	2	3
y	7.5	2	-1.5	-3	-2.5	0	4.5

Finish: Missing values are $2, -1.5, -3, 0$.

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 15 (a) Complete the table of values for y x x = - - 2 2 3 x -3 -2 -1 0 1 2 3 y 7.5 -2.5 4.5 (2) (b) On the grid, draw the graph of y x x = - - 2 2 3 for values of x from -3 to 3 -1 -2 O -1 -2 -3 -4 1 2 4 3 2 1 y x 3 -3 5 6 7 8 (2)

YOUR SOLUTION

02 Plot the curve

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 4(x-1)^2 - a$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q20

Calculate value

$$y = 4(x-1)^2 - a$$

The turning point is read

$$(1, -a)$$

Finish: turning point $(1, -a)$, y-intercept $(0, 4-a)$, x-intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 20 The curve C has equation y = 4(x - 1)2 - a where a > 4 Using the axes below, sketch the curve C. On your sketch show clearly, in terms of a, (i) the coordinates of any points of intersection of C with the coordinate axes, (ii) the coordinates of the turning point. y x O

YOUR SOLUTION

03 Read roots and intersections

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = -\frac{2}{x^2}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q15

Solve equation

$$y = -\frac{2}{x^2}$$

This is Graph D
This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: D, B, A.

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 15 Here are six graphs. Graph A y x O Graph B y x O Graph C y x O Graph D y x O Graph E y x O Graph F y x O Complete the table below with the letter of the graph that could represent each given equation. Write your answers on the dotted lines. Equation Graph y = 2 2 x y = - 1 2 x3 y = - 5 x

YOUR SOLUTION

04 Transform the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 4(x-1)^2 - a$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q20

Calculate value

$$y = 4(x-1)^2 - a$$

The turning point is read

$$(1, -a)$$

Finish: turning point $(1, -a)$, y-intercept $(0, 4-a)$, x-intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 20 The curve C has equation y = 4(x - 1)2 - a where a > 4 Using the axes below, sketch the curve C. On your sketch show clearly, in terms of a, (i) the coordinates of any points of intersection of C with the coordinate axes, (ii) the coordinates of the turning point. y x O

YOUR SOLUTION

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2x^2 - 4x - 1 = 0$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q21

Read the graph

$$2x^2 - 4x - 1 = 0$$

Read the graph

$$x = -0.2, \quad x = 2.2$$

Finish: (a) $-0.2,\ 2.2$; (b) draw $y = 1 - 2x$, giving $-0.6,\ 1.6$

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 21 Part of the graph of y = 2x2 - 4x - 1 is shown on the grid. -2 O 2 4 -4 -1 1 2 3 y x (a) Use the graph to find estimates for the solutions of the equation 2x2 - 4x - 1 = 0 Give your solutions correct to one decimal place. (2) (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation x2 - x - 1 = 0 Show your working clearly. Give your solutions correct to one decimal place. (3)

YOUR SOLUTION

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphs of Functions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = a\cos bx^\circ + c$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q26

The curve is

$$y = a\cos bx^\circ + c$$

Use trigonometry

$$3$$

Finish: $\boxed{a = 2,\ b = 3,\ c = 1}$

FORMULA BANK

Table of values, plot points, join with the correct curve shape, then read roots/intersections from the graph.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> 26 Here is a sketch of the curve with equation y = a cos bx degrees + c where -90 < x < 450 450 360 270 90 O -90 180 -2 -1 1 2 3 4 y x Find the value of a, the value of b and the value of c a = b = c =

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Complete a table of values
"draw", "plot", "construct", a diagram/grid	Plot the curve
"work out", "show", "hence", a familiar exam phrase	Read roots and intersections
"work out", "show", "hence", a familiar exam phrase	Transform the graph
"work out", "show", "hence", a familiar exam phrase	Read the graph
"work out", "show", "hence", a familiar exam phrase	Use trigonometry

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Estimating Gradients

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ESTIMATING GRADIENTS · CH3 · L6

ENG. ESLAM AHMED

The Map

This lesson collapses Estimating Gradients into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Draw a tangent

TRIGGER *"draw", "plot", "construct", a diagram/grid*

02 Use a large gradient triangle

TRIGGER *"gradient", "tangent", "rate of change"*

03 Estimate average gradient

TRIGGER *"gradient", "tangent", "rate of change"*

04 Interpret the rate

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

06 Draw a tangent to the

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BANK EVIDENCE 3 website questions, 3 checked solutions, 3 local PDFs read.

01 Draw a tangent

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m \approx -0.6$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q18

Draw a tangent to the

This is a reasoning step: write one clear sentence, then continue the calculation.

A reasonable tangent gives an

$$m \approx -0.6$$

Finish: Gradient ≈ -0.6, tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

Gradient at a curve = gradient of tangent. Average gradient = $\frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$ P -4 -2 O 2 4 6 8 10 12 2 4 -2 The point P on the curve has x coordinate 2 (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$ (2) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$ (b) Use the graph to find the two possible values of k. , (2)

YOUR SOLUTION

02 Use a large gradient triangle

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m \approx -0.6$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q18

Draw a tangent to the

This is a reasoning step: write one clear sentence, then continue the calculation.

A reasonable tangent gives an

$$m \approx -0.6$$

Finish: Gradient ≈ -0.6, tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

Gradient at a curve = gradient of tangent. Average gradient = $\frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 18 The diagram shows the graph of y = f (x) for -4 < x < 12 P -4 -2 O 2 4 6 8 10 12 2 4 -2 The point P on the curve has x coordinate 2 (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form y = mx + c (2) The equation f (x) = k has exactly two different solutions for -4 < x < 12 (b) Use the graph to find the two possible values of k. , (2)

YOUR SOLUTION

03 Estimate average gradient

TRIGGER *"gradient", "tangent", "rate of change"*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m \approx -0.6$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q18

Draw a tangent to the

This is a reasoning step: write one clear sentence, then continue the calculation.

A reasonable tangent gives an

$$m \approx -0.6$$

Finish: Gradient ≈ -0.6, tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

Gradient at a curve = gradient of tangent. Average gradient = $\frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$ P -4 -2 O 2 4 6 8 10 12 2 4 -2 The point P on the curve has x coordinate 2 (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$ (2) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$ (b) Use the graph to find the two possible values of k. , (2)

YOUR SOLUTION

04 Interpret the rate

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m \approx -0.6$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q18

Draw a tangent to the

This is a reasoning step: write one clear sentence, then continue the calculation.

A reasonable tangent gives an

$$m \approx -0.6$$

Finish: Gradient ≈ -0.6, tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

Gradient at a curve = gradient of tangent. Average gradient = $\frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 18 The diagram shows the graph of y = f (x) for -4 < x < 12 P -4 -2 O 2 4 6 8 10 12 2 4 -2 The point P on the curve has x coordinate 2 (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form y = mx + c (2) The equation f (x) = k has exactly two different solutions for -4 < x < 12 (b) Use the graph to find the two possible values of k. , (2)

YOUR SOLUTION

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m \approx -0.6$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q18

Draw a tangent to the

This is a reasoning step: write one clear sentence, then continue the calculation.

A reasonable tangent gives an

$$m \approx -0.6$$

Finish: Gradient ≈ -0.6, tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

Gradient at a curve = gradient of tangent. Average gradient = $\frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$ P -4 -2 O 2 4 6 8 10 12 2 4 -2 The point P on the curve has x coordinate 2 (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$ (2) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$ (b) Use the graph to find the two possible values of k. , (2)

YOUR SOLUTION

06 Draw a tangent to the

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Estimating Gradients question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$m \approx -0.6$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q18

Draw a tangent to the

This is a reasoning step: write one clear sentence, then continue the calculation.

A reasonable tangent gives an

$$m \approx -0.6$$

Finish: Gradient ≈ -0.6, tangent $y \approx -0.6x + 3.6$, and $k \approx 3, -1$.

FORMULA BANK

Gradient at a curve = gradient of tangent. Average gradient = $\frac{\Delta y}{\Delta x}$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> 18 The diagram shows the graph of $y = f(x)$ for $-4 < x < 12$ P -4 -2 O 2 4 6 8 10 12 2 4 -2 The point P on the curve has x coordinate 2 (a) (i) Use the graph to find an estimate for the gradient of the curve at P. (3) (ii) Hence find an equation of the tangent to the curve at P. Give your answer in the form $y = mx + c$ (2) The equation $f(x) = k$ has exactly two different solutions for $-4 < x < 12$ (b) Use the graph to find the two possible values of k. , (2)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"draw", "plot", "construct", a diagram/grid	Draw a tangent
"gradient", "tangent", "rate of change"	Use a large gradient triangle
"gradient", "tangent", "rate of change"	Estimate average gradient
"work out", "show", "hence", a familiar exam phrase	Interpret the rate
"work out", "find", a missing value	Find the gradient
"draw", "plot", "construct", a diagram/grid	Draw a tangent to the

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Real-Life Graphs

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

REAL-LIFE GRAPHS · CH3 · L7

ENG. ESLAM AHMED

The Map

This lesson collapses Real-Life Graphs into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Complete a table of values

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Plot the curve

TRIGGER *"draw", "plot", "construct", a diagram/grid*

03 Read roots and intersections

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Transform the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 42 website questions, 42 checked solutions, 3 local PDFs read.

01 Complete a table of values

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Real-Life Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = x^2 - \frac{x}{2} - 3$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q15

Solve quadratic equation

$$y = x^2 - \frac{x}{2} - 3$$

The completed table is

x	-3	-2	-1	0	1	2	3
y	7.5	2	-1.5	-3	-2.5	0	4.5

Finish: Missing values are $2, -1.5, -3, 0$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 15 (a) Complete the table of values for y x x = - - 2 2 3 x -3 -2 -1 0 1 2 3 y 7.5 -2.5 4.5 (2)
> (b) On the grid, draw the graph of y x x = - - 2 2 3 for values of x from -3 to 3 -1 -2 O -1 -2 -3 -4 1 2 4 3 2 1 y x 3 -3 5 6 7 8 (2)

YOUR SOLUTION

02 Plot the curve

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Real-Life Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 4(x-1)^2 - a$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q20

Calculate value

$$y = 4(x-1)^2 - a$$

The turning point is read

$$(1, -a)$$

Finish: turning point $(1, -a)$, y-intercept $(0, 4-a)$, x-intercepts $\left(1-\frac{\sqrt{a}}{2}, 0\right)$ and $\left(1+\frac{\sqrt{a}}{2}, 0\right)$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 20 The curve C has equation y = 4(x - 1)2 - a where a > 4 Using the axes below, sketch the curve C. On your sketch show clearly, in terms of a, (i) the coordinates of any points of intersection of C with the coordinate axes, (ii) the coordinates of the turning point. y x O

YOUR SOLUTION

03 Read roots and intersections

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Real-Life Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = -\frac{2}{x^2}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q15

Solve equation

$$y = -\frac{2}{x^2}$$

This is Graph D
This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: D, B, A.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 15 Here are six graphs. Graph A y x O Graph B y x O Graph C y x O Graph D y x O Graph E y x O Graph F y x O Complete the table below with the letter of the graph that could represent each given equation. Write your answers on the dotted lines. Equation Graph y = 2 2 x y = - 1 2 x3 y = - 5 x

YOUR SOLUTION

04 Transform the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Real-Life Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 4(x-1)^2 - a$$

SIMPLEST STRATEGY

1 Underline the target: what is the question asking you to find?

2 Write the rule, theorem, or equation before any calculator work.

3 Substitute one value at a time and keep units/unknowns visible.

4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q20

Calculate value

$$y = 4(x-1)^2 - a$$

The turning point is read

$$(1, -a)$$

Finish: turning point $(1, -a)$, y-intercept $(0, 4-a)$, x-intercepts $\left(1 - \frac{\sqrt{a}}{2}, 0\right)$ and $\left(1 + \frac{\sqrt{a}}{2}, 0\right)$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 20 The curve C has equation y = 4(x - 1)2 - a where a > 4 Using the axes below, sketch the curve C. On your sketch show clearly, in terms of a, (i) the coordinates of any points of intersection of C with the coordinate axes, (ii) the coordinates of the turning point. y x O

YOUR SOLUTION

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Real-Life Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2x^2 - 4x - 1 = 0$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q21

Read the graph

$$2x^2 - 4x - 1 = 0$$

Read the graph

$$x = -0.2, \quad x = 2.2$$

Finish: (a) $-0.2,\ 2.2$; (b) draw $y = 1 - 2x$, giving $-0.6,\ 1.6$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 21 Part of the graph of y = 2x2 - 4x - 1 is shown on the grid. -2 O 2 4 -4 -1 1 2 3 y x (a) Use the graph to find estimates for the solutions of the equation 2x2 - 4x - 1 = 0 Give your solutions correct to one decimal place. (2) (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation x2 - x - 1 = 0 Show your working clearly. Give your solutions correct to one decimal place. (3)

YOUR SOLUTION

06 Solve equation

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Real-Life Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = -\frac{2}{x^2}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q15

Solve equation

$$y = -\frac{2}{x^2}$$

This is Graph D
This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: D, B, A.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> 15 Here are six graphs. Graph A y x O Graph B y x O Graph C y x O Graph D y x O Graph E y x O Graph F y x O Complete the table below with the letter of the graph that could represent each given equation. Write your answers on the dotted lines. Equation Graph y = 2 2 x y = - 1 2 x3 y = - 5 x

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Complete a table of values
"draw", "plot", "construct", a diagram/grid	Plot the curve
"work out", "show", "hence", a familiar exam phrase	Read roots and intersections
"work out", "show", "hence", a familiar exam phrase	Transform the graph
"work out", "show", "hence", a familiar exam phrase	Read the graph
"work out", "show", "hence", a familiar exam phrase	Solve equation

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Graphing Inequalities

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

GRAPHING INEQUALITIES · CH3 · L8

ENG. ESLAM AHMED

The Map

This lesson collapses Graphing Inequalities into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Draw the boundary line

TRIGGER *"draw", "plot", "construct", a diagram/grid*

02 Choose solid or dashed

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Test one point

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Shade the correct region

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 The required region is

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 19 website questions, 19 checked solutions, 3 local PDFs read.

01 Draw the boundary line

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x \leq 0$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q10

The shaded region is left

$$x \leq 0$$

It is below the line

$$y \leq -x + 4$$

Finish: $x \leq 0, y \leq -x + 4, y \geq \frac{1}{3}x - 2$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 10 The shaded region in the diagram is bounded by three lines. The equation of one of the lines is given. y x -6 -4 -2 O -6 -4 -2 6 4 2 2 4 6 y = 1 3 x - 2 Write down the three inequalities that define the shaded region.

YOUR SOLUTION

02 Choose solid or dashed

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2 < x \leq 4$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q11

Solve inequality

$$2 < x \leq 4$$

The integer values are

$$3, 4$$

Finish: (a) $3, 4$. (b) $y \geq x - 3$, $x + y \geq 5$, $y \leq 6$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 11 (a) Write down the integer values of x that satisfy the inequality $-2 < x < 4$ (2) The region R, shown shaded in the diagram, is bounded by three straight lines. Diagram NOT accurately drawn R O y $y = x - 3$ x $x + y = 5$ $y = 6$ (b) Write down the three inequalities that define the region R. (2)

YOUR SOLUTION

03 Test one point

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = -3x + 5$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2HR Q7

A straight line with gradient

$$y = -3x + 5$$

Use the inequalities

$$x < 6$$

Finish: (a) $y = -3x + 5$. (b) Shade $x < 6, y > 2, y \leq x + 1$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 7 (a) Write down an equation of the straight line with gradient -3 and which passes through the point with coordinates (0, 5) (2) (b) Show, by shading on the grid, the region defined by all three of the inequalities x < 6 y 2 y < x + 1 Label the region R O 2 4 6 8 10 10 8 6 4 2 x y (3)

YOUR SOLUTION

04 Shade the correct region

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x \leq 0$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q10

The shaded region is left

$$x \leq 0$$

It is below the line

$$y \leq -x + 4$$

Finish: $x \leq 0, y \leq -x + 4, y \geq \frac{1}{3}x - 2$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 10 The shaded region in the diagram is bounded by three lines. The equation of one of the lines is given. y x -6 -4 -2 O -6 -4 -2 6 4 2 2 4 6 y = 1 3 x - 2 Write down the three inequalities that define the shaded region.

YOUR SOLUTION

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$9 - 4x > 17$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q8

Read the graph

$$9 - 4x > 17$$

Read the graph

$$y \geq 2$$

Finish: (a) $x < -2$. (b) $y \geq 2$, $x \leq 6$, $y \leq x$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 8 (a) Solve $9 - 4x > 17$ (2) O 1 2 3 4 5 6 7 8 1 2 3 4 5 6 7 8 9 10 y x R (b) Write down the three inequalities that represent the shaded region R (3)

YOUR SOLUTION

06 The required region is

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Graphing Inequalities question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 1, \qquad x = 2, \qquad x + y = 7$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2H Q2

Draw the boundary lines

$$y = 1, \qquad x = 2, \qquad x + y = 7$$

The required region is

$$y \geq 1$$

Finish: Shade $y \geq 1$, $x \geq 2$, $x + y \leq 7$.

FORMULA BANK

Boundary line first. Solid for $\leq, \geq$, dashed for $<, >$. Test a point for shading.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> 2 (a) On the grid, draw and label with its equation the straight line with equation (i) y = 1 (ii) x = 2 (iii) x + y = 7 O 1 2 3 4 5 6 7 8 x y 8 7 6 5 4 3 2 1 (3) (b) Show, by shading on the grid, the region that satisfies all three of the inequalities y 1 x 2 x + y < 7 Label the region R. (1)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"draw", "plot", "construct", a diagram/grid	Draw the boundary line
"work out", "show", "hence", a familiar exam phrase	Choose solid or dashed
"work out", "show", "hence", a familiar exam phrase	Test one point
"work out", "show", "hence", a familiar exam phrase	Shade the correct region
"work out", "show", "hence", a familiar exam phrase	Read the graph
"work out", "show", "hence", a familiar exam phrase	The required region is

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Transformations of Graphs

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ENG. ESLAM AHMED

The Map

This lesson collapses Transformations of Graphs into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Shift up or down

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Shift left or right

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Stretch vertically

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Reflect or scale horizontally

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 38 website questions, 38 checked solutions, 3 local PDFs read.

01 Shift up or down

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 2.5$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q22

Read the graph

$$a = 2.5$$

Use trigonometry

$$x + b = 360$$

Finish: $a = 2.5,\ b = 300;\ (2, 5)$ and $(4, -2)$

FORMULA BANK

$f(x) + a$ shifts up, $f(x - a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 22 The graph of $y = a \cos(x + b)$ degrees for $0 < x < 360$ is drawn on the grid. y x 360 300 240 180 120 60 O -3 -2 -1 1 2 3 (a) Find the value of a and the value of b. a = b = (2) Another curve C has equation $y = f(x)$ The coordinates of the minimum point of C are (4, 5) (b) Write down the coordinates of the minimum point of the curve with equation (i) $y = f(2x)$ (,) (ii) $y = f(x) - 7$ (,) (2)

YOUR SOLUTION

02 Shift left or right

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 2.5$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q22

Read the graph

$$a = 2.5$$

Use trigonometry

$$x + b = 360$$

Finish: $a = 2.5,\ b = 300;\ (2, 5)$ and $(4, -2)$

FORMULA BANK

$f(x) + a$ shifts up, $f(x - a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 22 The graph of $y = a \cos(x + b)$ degrees for $0 < x < 360$ is drawn on the grid. y x 360 300 240 180 120 60 O -3 -2 -1 1 2 3 (a) Find the value of a and the value of b. a = b = (2) Another curve C has equation $y = f(x)$ The coordinates of the minimum point of C are (4, 5) (b) Write down the coordinates of the minimum point of the curve with equation (i) $y = f(2x)$ (,) (ii) $y = f(x) - 7$ (,) (2)

YOUR SOLUTION

03 Stretch vertically

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = f(x+4)$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q21

The maximum point of is

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate value

$$y = f(x+4)$$

Finish: $(0,6)$, $(2,6)$, and $y = (x-4)^2 + 3(x-4) + 10$

FORMULA BANK

$f(x)+a$ shifts up, $f(x-a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 21 The diagram shows a sketch of part of the curve with equation y = f(x) y x (4, 6) y = f(x) O There is one maximum point on this curve. The coordinates of this maximum point are (4, 6) (a) Write down the coordinates of the maximum point on the curve with equation (i) y = f(x + 4) (,) (ii) y = f(2x) (,) (2) The equation of a curve C is y = x2 + 3x + 4 The curve C is transformed to curve S under the translation 4 6 (b) Find an equation of curve S. You do not need to simplify the equation. (2)

YOUR SOLUTION

04 Reflect or scale horizontally

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = f(x+4)$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q21

The maximum point of is

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate value

$$y = f(x+4)$$

Finish: $(0,6)$, $(2,6)$, and $y = (x-4)^2 + 3(x-4) + 10$

FORMULA BANK

$f(x) + a$ shifts up, $f(x-a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 21 The diagram shows a sketch of part of the curve with equation y = f(x) y x (4, 6) y = f(x) O There is one maximum point on this curve. The coordinates of this maximum point are (4, 6) (a) Write down the coordinates of the maximum point on the curve with equation (i) y = f(x + 4) (,) (ii) y = f(2x) (,) (2) The equation of a curve C is y = x2 + 3x + 4 The curve C is transformed to curve S under the translation 4 6 (b) Find an equation of curve S. You do not need to simplify the equation. (2)

YOUR SOLUTION

05 Read the graph

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Transformations of Graphs question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 2.5$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q22

Read the graph

$$a = 2.5$$

Use trigonometry

$$x + b = 360$$

Finish: $a = 2.5,\ b = 300;\ (2, 5)$ and $(4, -2)$

FORMULA BANK

$f(x) + a$ shifts up, $f(x - a)$ shifts right, $af(x)$ stretches vertically, $f(ax)$ changes horizontally.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 22 The graph of $y = a\cos(x + b)$ degrees for $0 < x < 360$ is drawn on the grid. y x 360 300 240 180 120 60 O -3 -2 -1 1 2 3 (a) Find the value of a and the value of b. a = b = (2) Another curve C has equation $y = f(x)$ The coordinates of the minimum point of C are (4, 5) (b) Write down the coordinates of the minimum point of the curve with equation (i) $y = f(2x)$ (,) (ii) $y = f(x) - 7$ (,) (2)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Shift up or down
"work out", "show", "hence", a familiar exam phrase	Shift left or right
"work out", "show", "hence", a familiar exam phrase	Stretch vertically
"work out", "show", "hence", a familiar exam phrase	Reflect or scale horizontally
"work out", "show", "hence", a familiar exam phrase	Read the graph

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Differentiation

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

DIFFERENTIATION · CH3 · L10

ENG. ESLAM AHMED

The Map

This lesson collapses Differentiation into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Differentiate powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find gradient at a point

TRIGGER *"work out", "find", a missing value*

03 Find stationary points

TRIGGER *"work out", "find", a missing value*

04 Use derivative in a context

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Differentiate

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 40 website questions, 40 checked solutions, 3 local PDFs read.

01 Differentiate powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$V = (2x+5)(x+1)(3-x)$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q15

The volume is

$$V = (2x+5)(x+1)(3-x)$$

Expand brackets

$$(2x+5)(x+1) = 2x^2 + 7x + 5$$

Finish: $x = 1.47$ to 3 significant figures.

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative $= 0$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 15 (2x + 5) cm (x + 1) cm (3 - x) cm Diagram NOT accurately drawn The diagram shows a cuboid of volume V cm3 (a) Show that V = 15 + 16x - x2 - 2x3 (3) There is a value of x for which the volume of the cuboid is a maximum. (b) Find this value of x. Show your working clearly. Give your answer correct to 3 significant figures. x = (5)

YOUR SOLUTION

02 Find gradient at a point

TRIGGER *"work out", "find", a missing value*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = ax^3 + bx^2 - 12x + 6$$

SIMPLEST STRATEGY

1 Underline the target: what is the question asking you to find?

2 Write the rule, theorem, or equation before any calculator work.

3 Substitute one value at a time and keep units/unknowns visible.

4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q24

Calculate value

$$y = ax^3 + bx^2 - 12x + 6$$

Point lies on the curve

$$-6 = 8a + 4b - 24 + 6$$

Finish: $y = 33$

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative $= 0$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 24 The curve C has equation y = ax3 + bx2 - 12x + 6 where a and b are constants. The point A with coordinates (2, -6) lies on C The gradient of the curve at A is 16 Find the y coordinate of the point on the curve whose x coordinate is 3 Show clear algebraic working. y = O A O A S 100 A S

YOUR SOLUTION

03 Find stationary points

TRIGGER *"work out", "find", a missing value*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = x^3 - 8x^2 - 12x + 5$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

> **Source pattern:** May 2024 P1HR Q19
>
> **Differentiate**
>
> $$y = x^3 - 8x^2 - 12x + 5$$
>
> **Stationary points have**
>
> $$3x^2 - 16x - 12 = 0$$
>
> **Finish:** $A = (6, -139)$.

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative $= 0$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 19 A curve C has equation y x x x 3 2 8 12 5 Curve C has exactly two stationary points, one at point A and one at point B such that x coordinate of point A > x coordinate of point B Find the coordinates of point A Show clear algebraic working. (,)

YOUR SOLUTION

04 Use derivative in a context

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q17

Volume of the solid is

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

The question gives

$$V = \frac{1}{3}\pi y$$

Finish: The maximum value of y is 320.

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative $= 0$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 17 A solid, S, is made from a hemisphere and a cylinder. The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point. x cm (20 - 4x) cm Diagram NOT accurately drawn The radius of the cylinder and the radius of the hemisphere are both x cm. The height of the cylinder is (20 - 4x) cm. The volume of S is V cm 3 where V = 1 3 piy Find the maximum value of y. Show clear algebraic working.

YOUR SOLUTION

05 Differentiate

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q17

Volume of the solid is

$$V = \pi x^2(20 - 4x) + \frac{2}{3}\pi x^3$$

The question gives

$$V = \frac{1}{3}\pi y$$

Finish: The maximum value of y is 320.

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative $= 0$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 17 A solid, S, is made from a hemisphere and a cylinder. The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point. x cm (20 - 4x) cm Diagram NOT accurately drawn The radius of the cylinder and the radius of the hemisphere are both x cm. The height of the cylinder is (20 - 4x) cm. The volume of S is V cm 3 where V = 1 3 piy Find the maximum value of y. Show clear algebraic working.

YOUR SOLUTION

06 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Differentiation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = x^2 - \frac{p}{x}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q22

Find the gradient

$$y = x^2 - \frac{p}{x}$$

Find the gradient

$$2(-3) + \frac{p}{(-3)^2} = 0$$

Finish: $p = 54,\ k = 12$

FORMULA BANK

If $y = ax^n$, then $\frac{dy}{dx} = anx^{n-1}$. Stationary point when derivative $= 0$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> 22 The diagram shows a sketch of part of the curve with equation y = x 2 - p x where p is a positive constant. Diagram NOT accurately drawn y x T O For all values of p, the curve has exactly one turning point and this turning point is a minimum shown as the point T in the sketch. For the curve where the x coordinate of T is -3 (a) find the value of p p = (4) The line with equation y = k is a tangent to the curve with equation y = x 2 - 16 x (b) Find the value of k k = (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Differentiate powers
"work out", "find", a missing value	Find gradient at a point
"work out", "find", a missing value	Find stationary points
"work out", "show", "hence", a familiar exam phrase	Use derivative in a context
"work out", "show", "hence", a familiar exam phrase	Differentiate
"work out", "find", a missing value	Find the gradient

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.